MILITARY SPECIFICATION

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM CATHODE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for tantalum cathode, electrolytic (nonsolid electrolyte), fixed capacitors, nonhermetically-sealed in metal cases (see 6.1).

1.2 Classification. Capacitors covered by this specification are classified by the style, as specified (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	-	Solder; Tin Alloy; Tin-Lead Alloy; and Lead Alloy.

MILITARY

MIL-C-39028	-	Capacitors, Packaging of.
MIL-C-45662	-	Calibration System Requirements.
MIL-C-83500/1	-	Capacitors, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum Cathode, Styles CRL01 and CRL02.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern (see 6.2).

3.2 Qualification. Capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Solder. Solder shall be in accordance with QQ-S-571.

3.3.2 Soldering flux. Soldering flux shall be rosin or rosin and alcohol; however, other soldering flux may be used if adequate evidence is submitted to the qualifying activity to indicate that the proposed flux is equally suitable.

3.4 Design and construction. Capacitors shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Case. Each capacitor shall be enclosed in a nonhermetically-sealed metal case (see 3.1), which will protect the capacitor element against the entry of contaminants.

3.4.2 Capacitor element. Both the anode and the cathode shall be of sintered tantalum with tantalum pentoxide dielectric.

3.4.3 Terminals. All terminals shall be permanently secured internally and externally, as applicable. All external joints shall be soldered or welded.

3.4.3.1 Radial pin. Radial pin terminals shall be hot-solder-dipped or electroplated with a solder having a maximum tin content of 40 to 70 percent. Radial pin terminals shall be solderable over their entire length up to 0.062 inch of the welded joint or up to 0.156 inch of the body, as applicable (see 3.1). Radial pin terminals shall be of a solid conductor of the length and diameter specified (see 3.1).

3.4.3.2 Stud. Stud terminals shall be threaded in accordance with Handbook H28 as specified (see 3.1).

3.5 Burn-in. When capacitors are tested as specified in 4.6.2, there shall be no mechanical damage or leakage of electrolyte.

3.6 DC leakage. When measured as specified in 4.6.3, the dc leakage shall not exceed the applicable value specified (see 3.1).

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3.7 Capacitance. When measured as specified in 4.6.4, the capacitance shall be within tolerance of the nominal value specified (see 3.1).

3.8 Dissipation factor. When capacitors are tested as specified in 4.6.5, the dissipation factor shall not exceed the applicable percent specified (see 3.1).

3.9 Shock (specified pulse). When capacitors are tested as specified in 4.6.6, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration, arcing, open- or short-circuiting, mechanical damage, or leakage of electrolyte.

3.10 Vibration, high frequency. When tested as specified in 4.6.7, there shall be no intermittent contacts of 0.5 ms or greater duration, open- or short-circuiting, and capacitors shall meet the following requirements:

DC leakage	Shall not exceed 125 percent of the initial requirement (see 3.6).
Capacitance	Shall change not more than ±5 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed 115 percent of the initial requirement (see 3.8).
Visual examination	There shall be no evidence of mechanical damage, obliteration of marking, or leakage of electrolyte.

3.11 Thermal shock. When tested as specified in 4.6.8, capacitors shall meet the following requirements:

DC leakage	Shall not exceed 125 percent of the initial requirement (see 3.6).
Capacitance	Shall change not more than ±5 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed 115 percent of the initial requirement (see 3.8).
Visual examination	There shall be no evidence of mechanical damage, obliteration of marking, or leakage of electrolyte.

3.12 Salt spray (corrosion). When capacitors are tested as specified in 4.6.9, there shall be no harmful corrosion, and at least 90 percent of any exposed metal surface shall be protected by the finish. Marking shall remain legible.

NOTE: Harmful corrosion shall be construed as being any type of corrosion which in any way interferes with the mechanical or electrical performance of the capacitor.

3.13 Solderability (capacitors with radial pin terminals and terminal tabs only, see 3.1). When capacitors are tested as specified in 4.6.10, the dipped portion of the terminals and terminal tabs shall conform to the solid-wire termination criteria of method 208 of MIL-STD-202.

3.14 Terminal strength. When capacitors are tested as specified in 4.6.11, there shall be no loosening or permanent damage to the terminals, terminal weld, or seal.

3.15 Surge voltage. When capacitors are tested as specified in 4.6.12, there shall be no intermittent contacts, open- or short-circuiting, mechanical damage, or leakage of electrolyte.

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3.16 Moisture resistance. When tested as specified in 4.6.13, capacitors shall meet the following requirements:

DC leakage	Shall not exceed 125 percent of the initial requirement (see 3.6).
Capacitance	Shall change not more than ±8 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed 115 percent of the initial requirement (see 3.8).
Visual examination	There shall be no evidence of harmful corrosion (as defined in 3.12), mechanical damage, obliteration of marking, or leakage of electrolyte.

3.17 Low temperature (storage). When tested as specified in 4.6.14, capacitors shall meet the following requirements:

DC leakage	Shall not exceed the initial requirement (see 3.6).
Capacitance	Shall change not more than ±5 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed the initial requirement (see 3.8).
External examination	There shall be no evidence of leakage of electrolyte.

3.18 Seal. When capacitors are tested as specified in 4.6.15, there shall be no evidence of leakage.

3.19 Stability at low and high temperatures. When tested as specified in 4.6.16, capacitors shall meet the following requirements:

Step 1 (+25°C):

DC leakage	Shall not exceed the initial requirement (see 3.6).
Capacitance	Shall be within the applicable tolerance specified (see 3.1).
Dissipation factor	Shall not exceed the initial requirement (see 3.8).

Step 2 (-55°C):

Impedance	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than the percent specified (see 3.1) from the step 1 measured value.

Step 3 (+25°C):

DC leakage	Shall not exceed the initial requirement (see 3.6).
Capacitance	Shall change not more than ±5 percent from the step 1 measured value.
Dissipation factor	Shall not exceed the initial requirement (see 3.8).

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Step 4 (+85°C):

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than the percent specified (see 3.1) from the step 1 measured value.
Dissipation factor	Shall not exceed the applicable value specified (see 3.1).

Step 5(+125°C):

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than the percent specified (see 3.1) from the step 1 measured value.
Dissipation factor	Shall not exceed the applicable value specified (see 3.1).

Step 6 (+25°C):

DC leakage	Shall not exceed the initial requirement (see 3.6).
Capacitance	Shall change not more than ±5 percent from the step 1 measured value.
Dissipation factor	Shall not exceed the initial requirement (see 3.8).

3.20 Barometric pressure (reduced). When capacitors are tested as specified in 4.6.17, there shall be no flashover, breakdown, or harmful deformation of the case, and no mechanical damage, obliteration of marking, or leakage of electrolyte.

3.21 Reverse voltage (applicable to polarized capacitors only, see 3.1). When tested as specified in 4.6.18, capacitors shall meet the following requirements:

DC leakage	Shall not exceed 125 percent of the initial requirement (see 3.6).
Capacitance	Shall change not more than ±10 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed the initial requirement (see 3.8).

3.22 Life at 85°C.

3.22.1 2,000-hour (qualification inspection). When tested as specified in 4.6.19.1, capacitors shall meet the following requirements:

DC leakage (at 85°C and 25°C)	Shall not exceed the applicable values specified (see 3.1).
Capacitance	Shall change not more than ±10 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed the initial requirement (see 3.8).
Visual examination	There shall be no mechanical damage, obliteration of marking, or leakage of electrolyte.

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3.22.2 240-hour (group B inspection). When tested as specified in 4.6.19.2, capacitors shall meet the following requirements:

DC leakage (at 85°C and 25°C)	Shall not exceed the applicable values specified (see 3.1).
Capacitance	Shall change not more than 10 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed the initial requirement (see 3.8).
Visual examination	There shall be no mechanical damage, obliteration of marking, or leakage of electrolyte.

3.22.3 6,000- and 10,000-hour (group C inspection). When tested as specified in 4.6.19.3, capacitors shall meet the following requirements:

DC leakage:	
At 85°C	Shall not exceed 125 percent of the initial requirement (see 3.6).
At 25°C	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than the percent specified (see 3.1) from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed 150 percent of the initial requirements (see 3.8).

3.23 Life at 125°C. When tested as specified in 4.6.20, capacitors shall meet the following requirements:

DC leakage (at 125°C and 25°C)	Shall not exceed the applicable values specified (see 3.1).
Capacitance	Shall change not more than ±10 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed 150 percent of the initial requirement (see 3.8).
Visual examination	There shall be no damage, obliteration of marking, or leakage of electrolyte.

3.24 Vacuum life at 125°C. When tested as specified in 4.6.21, capacitors shall meet the following requirements:

DC leakage (at 125°C and 25°C)	Shall not exceed the applicable values specified (see 3.1).
Capacitance	Shall change not more than ±20 percent from the initial measured value (see 3.7).
Dissipation factor	Shall not exceed the initial requirement (see 3.8).
Weight loss	Shall not exceed 4.1 milligrams.

3.25 AC ripple life at 125°C. When tested as specified in 4.6.22, capacitors shall meet the following requirements:

DC leakage (at 125°C and 25°C)	Shall not exceed the applicable values specified (see 3.1).

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Capacitance - - - - - - - - - -	Shall change not more than ±15 percent from the initial measured value (see 3.7).
Dissipation factor- - - - - - -	Shall not exceed the applicable value specified (see 3.1).
Visual examination - - - - - -	There shall be no damage, obliteration of marking, or leakage of electrolyte.

3.26 Fungus. The manufacturer shall certify that all external materials are fungus resistant or shall perform the test specified in 4.6.23. When capacitors are tested as specified in 4.6.23, examination shall disclose no evidence of fungus growth on the external surface of the capacitor.

3.27 Resistance to solvents. When capacitors are tested as specified in 4.6.24, marking shall remain legible and shall not smear or rub off. In addition, there shall be no visible indication of damage or deterioration to the capacitor body.

3.28 Resistance to soldering heat (capacitors with terminal tabs only). When tested as specified in 4.6.25, capacitors shall meet the following requirements:

DC leakage - - - - - - - - - -	Shall not exceed the initial requirement (see 3.6).
Capacitance - - - - - - - - -	Shall change not more than ±5 percent from the initial measured value (see 3.7).
Dissipation factor - - - - -	Shall not exceed the initial requirement (see 3.8).

3.29 Marking. Capacitors shall be marked in accordance with method I of MIL-STD-1285. Unless otherwise specified (see 3.1), marking shall include the Military part number, capacitance (in µF), and rated voltage (at 85°C) as shown in the following example:

EXAMPLE:

M83500
01-0001 - Military part number.
1200µF 6V - Capacitance and 85°C rated voltage.

When applicable, polarity marking shall be as specified (see 3.1). The Military part number, capacitance, rated voltage, source code, date code, lot symbol, and trademark shall be marked on the packaging container.

3.30 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality and shall be free from misalined welds, cold soldering, harmful corrosion (as defined in 3.12), pits, cracks, dents, rough edges, and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions and methods.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 Methods.

4.3.2.1 AC measurements. AC measurements shall be made at a frequency of 120 ±5 Hertz (Hz). The magnitude of the ac voltage shall be equal to or less than 1.0 volt root mean square (rms). The dc bias voltage shall be 2.2 $^{+0.0}_{-1.0}$ volts for all ac measurements. The ratio of ac voltage to dc bias voltage shall be such that no reversal of dc voltage will occur across the capacitor.

4.3.2.2 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at 25° ±5°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.3 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the applicable applied test voltage. The power supply used for dc leakage current measurements shall be stabilized to at least ±100 parts per million. No voltage fluctuations shall occur during measurements that would produce a variation in the current measurement.

4.4 Qualification inspection.1/ Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. The number of capacitors to be subjected to qualification inspection shall be as specified in table I, or in the appendix to this specification.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown. Two sample units shall be subjected to the visual and mechanical examination (internal). The remaining sample units shall be subjected to the subsequent inspections of group I. The sample shall then be divided as specified in table I for groups II to VII inclusive, and subjected to the inspections for their particular group.

4.4.3 Failures. Failures in excess of those allowed in table I shall be cause for refusal to grant qualification approval.

1/ Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

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TABLE I. Qualification inspection.

Examination or test	Requirement paragraph	Test paragraph	Number of sample units to be inspected 1/	Number of failures permitted 2/
Group I				
Visual and mechanical examination (internal): Materials, design, construction, and workmanship	3.1, 3.3, 3.4, and 3.30	4.6.1	2	0
Visual and mechanical examination (external): 3/ Physical dimensions, materials, marking4/, and workmanship	3.1, 3.3, 3.29, and 3.30	4.6.1	70	1
Burn-in 3/	3.5	4.6.2		
DC leakage 3/	3.6	4.6.3		
Capacitance 3/	3.7	4.6.4		
Dissipation factor 3/	3.8	4.6.5		
Group II				
Shock (specified pulse)	3.9	4.6.6	6	1
Vibration, high frequency	3.10	4.6.7		
Thermal shock	3.11	4.6.8		
Salt spray (corrosion)	3.12	4.6.9		
Group III				
Solderability (capacitors with radial pin terminals and terminal tabs only)	3.13	4.6.10	6	2
Terminal strength	3.14	4.6.11		1
Surge voltage	3.15	4.6.12		
Moisture resistance	3.16	4.6.13		
Low temperature (storage)	3.17	4.6.14		
Seal	3.18	4.6.15		
Group IV				
Stability at low and high temperatures	3.19	4.6.16	6	1
Barometric pressure (reduced)	3.20	4.6.17		
Reverse voltage (applicable to polarized capacitors only)	3.21	4.6.18		
Group V				
Life at 85°C (2,000-hour)	3.22.1	4.6.19.1	12	1
Life at 125°C (2,000-hour)	3.23	4.6.20	6	

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TABLE I. Qualification inspection. -Continued.

Examination or test	Requirement paragraph	Test paragraph	Number of sample units to be inspected 1/	Number of failures permitted 2/
Group VI				
Vacuum life at 125°C	3.24	4.6.21	27	1
AC ripple life at 125°C	3.25	4.6.22		
Group VII				
Fungus	3.26	4.6.23	6	1
Resistance to solvents	3.27	4.6.24		
Resistance to soldering heat (capacitors with terminal tabs only)	3.28	4.6.25		

1/ One additional sample unit is included in each sample of 72 units to permit substitution for the failure allowed in group I.
2/ A sample unit having one or more defects shall be considered as a single failure.
3/ Nondestructive examination and tests.
4/ Marking defects are based on visual examination only and shall be charged only for illegible, incomplete, or incorrect marking.

4.4.4 Retention of qualification. To retain qualification, the supplier shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, groups A and B, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
(b) The results of tests performed for qualification verification inspection, group C, including the number and mode of failures. The test report shall include results of all qualification verification inspection tests performed and completed during the 12-month period. If the test results indicate nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each style to testing in accordance with the qualification inspection requirements.

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4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all capacitors from a production line or lines, produced under essentially the same conditions and offered for inspection during a single work-week.

4.5.1.2 Group A inspection. Group A inspection shall consist of the examination and tests specified in table II, in the order shown.

TABLE II. Group A inspection.

Examination or test	Requirement paragraph	Test paragraph	AQL (percent defective)	
			Major	Minor
Burn-in- - - - - - - - - - - - - - -	3.5	4.6.2	Not applicable (100% inspection)	
Visual and mechanical examination (external):				
Materials- - - - - - - - - - - -	3.1 and 3.3	4.6.1	0.65	1.5
Physical dimensions- - - - - -	3.1	4.6.1		
Marking 1/ - - - - - - - - - - -	3.29	4.6.1		
Workmanship - - - - - - - - - -	3.30	4.6.1		
DC leakage- - - - - - - - - - - -	3.6	4.6.3	Not applicable (100% inspection)	
Capacitance- - - - - - - - - - - -	3.7	4.6.4		
Dissipation factor - - - - - - - -	3.8	4.6.5		

1/ Marking defects shall be based on visual examination only. Any subsequent electrical defects shall not be used as a basis for determining marking defects.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table II. Major and minor defects shall be as defined in MIL-STD-105 and as specified in table II.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table III, in the order shown, and shall be performed on sample units which have been subjected to and have passed group A inspection.

TABLE III. Group B inspection.

Test	Requirement paragraph	Test paragraph
Subgroup 1		
Stability at low and high temperatures - - - - - - - - - - - - -	3.19	4.6.16
Subgroup 2		
Life at 85°C (240-hour)- - - - - - - -	3.22.2	4.6.19.2

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4.5.1.3.1 Sampling plan. For subgroup 1, the sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4; (the sample size shall be based on the inspection lot size from which the sample was selected for group A inspection); the AQL shall be 1.5 percent defective. For subgroup 2, 10 samples shall be selected from units which have passed group A inspection; one failure is permitted.

4.5.1.3.2 Failure in group B inspection. If an inspection lot is rejected under the subgroup 1 test of table III, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots. If an inspection lot is rejected under the subgroup 2 test, the lot shall not be resubmitted.

4.5.1.3.3 Disposition of sample units. Sample units which have been subjected to subgroup 1 of group B inspection shall not be delivered on the contract or purchase order. Sample units which have been subjected to and have successfully passed subgroup 2 of group B inspection may be delivered on the contract or purchase order; however, every four months, 10 samples which have successfully passed the test of subgroup 2 shall be tested as specified in subgroup 6 of table IV.

4.5.2 Qualification verification inspection. Qualification verification inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.2.1.4), delivery of products which have passed groups A and B shall not be delayed pending the results of these qualification verification inspections.

4.5.2.1 Group C inspection. Group C inspection shall consist of the tests specified in table IV, in the order shown. Group C inspection shall be made on sample units which have passed group A inspection, except that units which are to be subjected to the tests of subgroup 6 shall be units which have successfully passed subgroup 2 of table III (see 4.5.1.3.3).

4.5.2.1.1 Sampling plan. Sample units shall be selected from the first lot and thereafter as specified in table IV. The voltages selected shall be represented in the sample in the same proportion as the voltages produced during the sampling period.

4.5.2.1.2 Failures. If the number of failures exceeds the number allowed in table IV, the sample shall be considered to have failed.

4.5.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.5.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.3 Inspection of preparation for delivery. Sample packages and packs shall be selected and inspected in accordance with MIL-C-39028 to verify conformance with the requirements in section 5 of this specification.

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TABLE IV. Group C inspection.

Test	Requirement paragraph	Test paragraph	Number of sample units to be inspected	Number of failures permitted 1/
Subgroup 1(every 2 months)				
Shock (specified pulse)- - - - -	3.9	4.6.6	12	1
Vibration, high frequency- - - -	3.10	4.6.7		
Salt spray (corrosion)- - - - -	3.12	4.6.9		
Subgroup 2 (every 2 months)				
Solderability (capacitors with radial pin terminals and terminal tabs only)- - - - - -	3.13	4.6.10	12	
Terminal strength- - - - - - - -	3.14	4.6.11		
Surge voltage- - - - - - - - -	3.15	4.6.12		
Moisture resistance- - - - - - -	3.16	4.6.13		
Seal- - - - - - - - - - - - - -	3.18	4.6.15		
Subgroup 3(every 2 months)				
Reverse voltage (applicable to polarized capacitors only) - -	3.21	4.6.18	6	
Subgroup 4(every 2 months)				
Low temperature (storage) - - -	3.17	4.6.14	12	1
Subgroup 5 (every 4 months)				
Thermal shock- - - - - - - - -	3.11	4.6.8	12	1
Subgroup 6(every 4 months)				
Life at 85°C - - - - - - - - - -	3.22.3	4.6.19.3	6	1
Barometric pressure (reduced) -	3.20	4.6.17		
Subgroup 7(every 4 months)				
Life at 125°C - - - - - - - -	3.23	4.6.20	6 2/	1
Subgroup 8(every 12 months)				
Vacuum life at 125°C - - - - -	3.24	4.6.21	27	1
AC ripple life at 125°C - - - - -	3.25	4.6.22		
Subgroup 9(every 12 months)				
Fungus- - - - - - - - - - - - -	3.26	4.6.23	6	1
Resistance to solvents- - - - -	3.27	4.6.24		
Resistance to soldering heat (capacitors with terminal tabs only)- - - - - - - - -	3.28	4.6.25		

1/ A sample unit having one or more defects shall be considered as a single failure.
2/ Sample units may be selected from units successfully passing subgroup 3.

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4.6 Methods of examination and test.

4.6.1 *Visual and mechanical examination.* Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.29, and 3.30).

4.6.2 Burn-in (see 3.5). Capacitors shall be subjected to five cycles of thermal shock as specified in 4.6.8. Capacitors shall be conditioned at 85°C with the applicable rated voltage applied through a 1,000 ±100 ohm resistor for a period of 48 $^{+5}_{-0}$ hours. Capacitors shall be returned to room ambient temperature and shall then be visually examined for mechanical damage and leakage of electrolyte.

NOTE: For nonpolarized capacitors, the conditioning voltage shall be reversed periodically so that the time of application of applicable voltages in both directions is approximately equal.

4.6.3 DC leakage (see 3.6). DC leakage shall be measured with the dc rated voltage (see 3.1) applied. The electrification period shall be as follows:

(a) For qualification inspection - 5 ±1/2 minutes.
(b) For quality conformance inspection - Within 5 minutes.

4.6.4 Capacitance (see 3.7). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. The following details shall apply:

(a) Test frequency - As specified in 4.3.2.1.
(b) Limit of accuracy - Within ±2 percent.

4.6.5 Dissipation factor (see 3.8). The dissipation factor shall be determined by a polarized bridge. Measurement accuracy shall be within ±2 percent.

4.6.6 Shock (specified pulse)(see 3.9). Capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting method - Normal mounting means.
(b) Test condition - I (100 G).
(c) Measurement and electrical loading during test - DC rated voltage (see 3.1) shall be applied to the capacitor during test. Observation shall be made to determine intermittent contacts or arcing or open- or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption of 0.5 ms or greater duration.
(d) Examination after test - Capacitors shall be visually examined for arcing, breakdown, mechanical damage, and leakage of electrolyte.

4.6.7 Vibration, high frequency (see 3.10). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exception shall apply:

(a) Mounting of specimens - Normal mounting means.
(b) Test condition - D (20 G), with the following exception:
For capacitors with radial pin terminals, the motion shall be applied for a total of 8 hours (4 hours in each of two mutually perpendicular directions, one parallel and the other perpendicular to the capacitor body).
(c) Measurements - During the last 30 minutes of vibration in each direction, an electrical measurement shall be made on the capacitors to determine intermittent contacts or open- or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption of 0.5 ms or greater duration.

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(d) Measurements after cycling - DC leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively. In addition, capacitors shall be externally examined for leakage of electrolyte. Detection of leakage of electrolyte shall be determined by the use of thymol blue or other suitable chemical indicator solution.
(e) Examination after test - Capacitors shall be visually examined for mechanical damage and leakage of electrolyte.

4.6.8 Thermal shock (see 3.11). Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

(a) Conditioning prior to first cycle - 15 minutes at room ambient conditions (see 4.3.1).
(b) Test condition - A (except step 3 temperature shall be $+125^{\circ}{}^{+4^{\circ}}_{-0^{\circ}}$C; steps 1 and 3 time shall be 30 minutes; and steps 2 and 4 time shall be 10 to 15 minutes).
(c) Number of cycles - 300.
(d) Measurements before cycling - Not applicable.
(e) Measurements after cycling - DC leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively. In addition, capacitors shall be externally examined for leakage of electrolyte. Detection of leakage of electrolyte shall be determined by the use of thymol blue or other suitable chemical indicator solution.

4.6.9 Salt spray (corrosion)(see 3.12). Capacitors shall be tested in accordance with method 101 of MIL-STD-202. The following details and exception shall apply:

(a) Test condition - B (48 hours).
(b) Measurements after exposure - Not applicable.
(c) Examination after test - After the test, capacitors shall be washed, and then air-dried for 24 hours. The capacitors shall then be examined for evidence of harmful corrosion, loss of plating, and legibility of marking.

4.6.10 Solderability (capacitors with radial pin terminals and terminal tabs only, see 3.1) (see 3.13). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

(a) Number of terminations of each part to be tested - 5 (CRL01) or 1 (CRL02).
(b) Depth of immersion - Terminals shall be immersed up to 0.062 inch of the welded joint or up to 0.156 inch of the body, whichever applies.

4.6.11 Terminal strength (see 3.14). Capacitors shall be tested in accordance with 4.6.11.1 through 4.6.11.3, as applicable.

4.6.11.1 Pull test (radial pin and stud terminals). Capacitors shall be tested in accordance with method 211 of MIL-STD-202. The following details and exception shall apply:

(a) Test condition - A.
(b) Applied force - 10 pounds, $^{+2}_{-0}$ ounces.
(c) Duration of applied force - 30 $^{+5}_{-0}$ seconds.

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4.6.11.2 Bend test (radial pin terminals and capacitors with terminal tabs only). Capacitors shall be tested in accordance with method 211 of *MIL-STD-202*. The following detail and exception shall apply:

(a) Test condition - B.
(b) Number of bends - Two, 90 degrees each for terminal tabs; or two, 45 degrees each for radial pin terminals. Capacitors with radial pin terminals shall be secured in position as shown in figure 1.

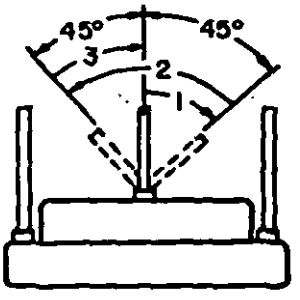


FIGURE 1. Wire-lead bend test.

4.6.11.3 Torque test (stud terminals only). Capacitors shall be tested in accordance with method 211 of *MIL-STD-202*. The following details shall apply:

(a) Test condition - E.
(b) Torque to be applied - 11 pound-inches.

After the tests, the terminals, terminal weld, and seals shall be examined for loosening or permanent damage.

4.6.12 Surge voltage (see 3.15). Capacitors shall be subjected to 1,000 cycles of the applicable dc surge voltage specified in table V. The ambient temperature during cycling shall be 85°C. Each cycle shall consist of a 30-second surge voltage application followed by a 5-1/2 minute discharge period. Voltage application shall be made through a resistance of 10^{+0}_{-2} ohms including the source in series with the capacitor and the voltage source. Each surge voltage cycle shall be performed in such a manner that the capacitor is discharged through a 10^{+0}_{-2} ohm resistor at the end of the 30-second application. The test shall be terminated on the discharge portion of the cycle. During the test, an electrical measurement shall be made to detect intermittent contacts and open- and short-circuiting. After the test, capacitors shall be visually examined for mechanical damage and leakage of electrolyte.

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TABLE V. DC surge derated, and rated voltages.

DC surge voltage at 85°C	Derated voltage at 125°C	DC rated voltage at 85°C
143.8	94.9	125
115.0	75.9	100
86.3	56.9	75
57.5	37.8	50
34.5	22.8	30
23.0	15.2	20
17.3	11.4	15
11.5	7.6	10
9.2	6.1	8
6.9	4.6	6

4.6.13 Moisture resistance (see 3.16). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - Except during examination and measurements, capacitors shall be securely fastened by normal mounting means.
(b) Initial measurements - Not applicable.
(c) Polarization and loading voltage - 6 volts dc or rated voltage, whichever is less.
(d) Step 7b - Vibration is not required during step 7b.
(e) Final measurements - After the final cycle and within 2 to 6 hours after removal of capacitors from the humidity chamber, the dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively.
(f) Examination after test - Capacitors shall be visually examined for evidence of harmful corrosion, mechanical damage, obliteration of marking, and leakage of electrolyte.

4.6.14 Low temperature (storage) (see 3.17). Capacitors shall be tested in accordance with method 502 of MIL-STD-810. The following details and exceptions shall apply:

(a) Pretest data required - DC leakage, capacitance, and dissipation factor measurements obtained from 4.6.3, 4.6.4, and 4.6.5, respectively.
(b) Storage temperature and exposure time - $-62^{\circ}{}^{+0^{\circ}}_{-3^{\circ}}$C for 72 hours, followed by a 1-hour exposure at $+125^{\circ}{}^{+4^{\circ}}_{-0^{\circ}}$C within 24 hours after low temperature storage.
(c) Steps not required - Steps 3, 4, and 5.
(d) Measurements after exposure - Capacitors shall be returned to the inspection conditions specified in 4.3.1; and within 24 hours after exposure, dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively.
(e) Examination after test - Capacitors shall be externally examined for leakage of the electrolyte. Detection of leakage of electrolyte shall be determined by the use of thymol blue, or other suitable chemical-indicator solution.

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4.6.15 Seal (see 3.18). Capacitors shall be tested in accordance with method 112 of MIL-STD-202. The following details shall apply:

(a) Test condition - C.
(b) Procedure number and degree of leakage-rate sensitivity - Procedure III, 10^{-8} atm cc/sec. Upon completion of this procedure, test-condition A shall be used for checking gross leaks.
(c) Measurements after test - Not applicable.

Following this test, capacitors shall be visually examined for evidence of leakage.

4.6.16 Stability at low and high temperatures (see 3.19). The measurements specified in table VI shall be made in the order shown. Capacitors shall be brought to thermal stability before the measurements are made. Thermal stability will have been reached when no further change in capacitance is observed between two successive measurements taken at 15-minute intervals.

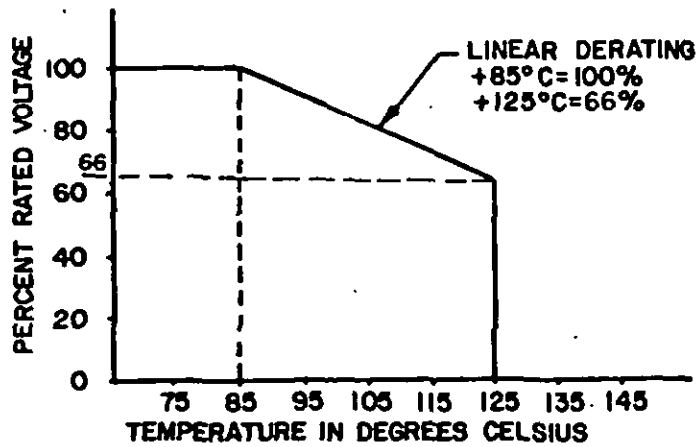
4.6.16.1 *Impedance.* Capacitors shall be conditioned at $-55^{\circ}{}^{+0^{\circ}}_{-3^{\circ}}$C for a period of not less than 30 minutes or until stable measurements can be obtained. The impedance shall then be measured directly or determined from measurements obtained on a bridge. Measurements shall be made at 120 ±5 Hz. Measurement accuracy shall be within ±5 percent.

TABLE VI. Temperatures and measurements for stability tests at low and high temperatures.

Step	Temperature	Measurement	Test paragraph
1	+25°C	DC leakage Capacitance Dissipation factor	4.6.3 4.6.4 4.6.5
2	$-55^{\circ}{}^{+0^{\circ}}_{-3^{\circ}}$C	Impedance Capacitance	4.6.16.1 4.6.4
3	+25°C	DC leakage Capacitance Dissipation factor	4.6.3 4.6.4 4.6.5
4	$+85^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}$C	DC leakage Capacitance Dissipation factor	4.6.3 4.6.4 4.6.5
5	$+125^{\circ}{}^{+4^{\circ}}_{-0^{\circ}}$C	DC leakage 1/ Capacitance Dissipation factor	4.6.3 4.6.4 4.6.5
6	+25°C	DC leakage Capacitance Dissipation factor	4.6.3 4.6.4 4.6.5

1/ During this measurement, derated voltage (see table V) shall be applied (see figure 2).

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NOTE: A polarizing voltage must be applied to the capacitor which on negative alternations of the ripple voltage will preclude polarity reversal and on positive alternations will not cause the capacitor rated voltage to be exceeded.

FIGURE 2. Voltage derating with temperature (85° to 125°C) (styles CRL01 and CRL02).

4.6.17 Barometric pressure (reduced) (see 3.20). Capacitors shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - By normal mounting means.
(b) Test condition - E (150,000 ft).
(c) Test during subjection to reduced pressure - At the end of the conditioning period and with the capacitors still at reduced pressure, the specified voltage (see 3.1) shall be applied for 1 minute ±5 seconds.
(d) Test after subjection to reduced pressure - Capacitors shall be visually examined for harmful deformation of the case, mechanical damage, obliteration of marking, leakage of electrolyte, and indications of flashover and breakdown.
(e) Exposure time prior to measurements - 5 minutes (minimum).

4.6.18 Reverse voltage (applicable to polarized capacitors only, see 3.1) (see 3.21). Capacitors shall be subjected to a dc potential of 3 volts, applied in the reverse polarity direction, for 125 ±10 hours. The ambient temperature during the test shall be 125°C. Capacitors shall be maintained at the 125°C temperature and derated voltage shall be applied in the forward polarity direction for an additional period of 125 ±10 hours. Capacitors shall then be returned to room ambient temperature and dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively.

4.6.19 Life at 85°C (see 3.22).

4.6.19.1 2,000-hour (qualification inspection)(see 3.22.1). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

(a) Distance of temperature measurements from specimens - Not applicable.
(b) Method of mounting - Normal mounting means.

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(c) Test temperature and tolerance - $+85^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}C$.

(d) Operating conditions - DC rated voltage (see 3.1) shall be applied gradually (not to exceed 5 minutes either by a slow build-up of the voltage or through a resistor which shall be shorted out within 5 minutes). Voltage shall be applied continuously except for measurement periods. The impedance of the voltage source, as seen from the terminals of each capacitor, shall not exceed 10 ohms. Storage batteries (or an electronic power supply) capable of supplying at least 100 milliamperes when a capacitor is shorted out shall be used. For nonpolarized capacitors, the voltage shall be reversed every 125 ±10 hours

(e) Test condition - F (2,000 $^{+72}_{-0}$ hours).

(f) Measurements during exposure (at +85°C) - DC leakage shall be measured as specified in 4.6.3 during the first hour of exposure and then at 240 $^{+48}_{-0}$; 1,000 $^{+48}_{-0}$; and 2,000 $^{+72}_{-0}$ hours.

(g) Measurements after exposure - Capacitors shall be returned to the inspection conditions specified in 4.3.1 and dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively. Capacitors shall then be visually examined for evidence of mechanical damage, obliteration of marking, and leakage of electrolyte.

4.6.19.2 240-hour (group B inspection) (see 3.22.2). Capacitors shall be tested as specified in 4.6.19.1, except that the duration of the test shall be 240 $^{+48}_{-0}$ hours.

4.6.19.3 6,000- and 10,000-hour (group C inspection) (see 3.22.3). Capacitors shall be tested as specified in 4.6.19.1, except as specified in the following:

(a) Test duration - 10,000 $^{+96}_{-0}$ hours.

(b) Measurements during exposure - DC leakage shall be measured during the first hour of exposure and then at 240 $^{+48}_{-0}$, 1,000 $^{+48}_{-0}$, 2,000 $^{+72}_{-0}$, and every 2,000 $^{+72}_{-0}$ hours thereafter until 10,000 $^{+96}_{-0}$ hours have elapsed.

(c) At the end of 6,000 $^{+72}_{-0}$ and 10,000 $^{+96}_{-0}$ hours, capacitors shall be returned to the inspection conditions specified in 4.3.1, and dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively. Capacitors shall then be visually examined for evidence of mechanical damage, obliteration of marking, and leakage of electrolyte.

4.6.20 Life at 125°C (see 3.23). Capacitors shall be tested as specified in 4.6.19.1 and 4.6.19.3, with the following exceptions:

(a) Test temperature and tolerance - $125^{\circ}{}^{+4^{\circ}}_{-0^{\circ}}C$.

(b) Operating conditions - Derated voltage (see table V) shall be applied.

4.6.21 Vacuum life at 125°C (see 3.24). Capacitors shall be conditioned at 125°C for a period of 50 hours maximum. Capacitors shall then be returned to the inspection conditions specified in 4.3.1 and each capacitor shall be weighed individually.

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Atmospheric pressure shall be reduced to $1x10^{-6}$ mm (±10 percent) and capacitors shall be tested as specified in 4.6.19.1(a) to 4.6.19.1(d) inclusive. Test duration shall be 500^{+48}_{-0} hours. Capacitors shall then be returned to the inspection conditions specified in 4.3.1 and dc leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively. Following these measurements, each capacitor shall be weighed individually. Weight measurement accuracy shall be ±0.05 milligram.

4.6.22 AC ripple life at 125°C (see 3.25). Capacitors shall be tested as specified in 4.6.19.1, with the following exception:

(a) Operating conditions - Capacitors shall be tested at 40 ±2 kHz sinusoidal and 750 mA rms (±10 percent). The peak ac voltage plus dc voltage shall not exceed the rated voltage of the capacitor.

4.6.23 Fungus (see 3.26). Unless certification is substituted, capacitors shall be tested in accordance with method 508 of MIL-STD-810. Pretest and post-test measurements are not required.

4.6.24 Resistance to solvents (see 3.27). Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) Portion of specimen to be brushed - That portion on which marking is present.
(b) Number of specimens to be tested - As specified in applicable inspection tables.
(c) Permissible extent of damage - As specified in 3.27.

4.6.25 Resistance to soldering heat(capacitors with terminal tabs only) (see 3.28). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

(a) Depth of immersion - Within 0.250-inch of the seal or case.
(b) Test-condition - B (260° ±5°C; 10 ±1 seconds).
(c) Cooling time prior to final measurements - 10 ±1 minutes.
(d) Measurements after test - DC leakage, capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, and 4.6.5, respectively.

5. PREPARATION FOR DELIVERY

5.1 Capacitors shall be prepared for delivery in accordance with MIL-C-39028. In addition to any special marking required by the contract or purchase order (see 6.2), each unit package shall be marked as specified in MIL-STD-1285.

6. NOTES

6.1 Intended use. These capacitors are used in "heavy duty" applications, such as aircraft engine controls, avionics, aerospace, etc. These capacitors are designed specifically to meet the unique needs of first and second stage filtering in static inverters, modulated regulators, and other heavy duty power supply applications including protection against reverse voltage.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the complete part number (see 3.1).

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6.3 *Qualification.* With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the qualifying activity below:

Commander
Air Force Logistics Command
Attention: MMGE
Wright-Patterson Air Force Base, Ohio 45433

6.3.1 Copies of "Provisions Governing Qualification (SD-6)" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 *Life degradation.* The life of these capacitors is primarily dependent upon voltage and temperature. When increased life is desired, capacitors may be further derated from the conditions specified in the specification. Under no condition should these capacitors be subjected to voltages above the derated voltage at the maximum rated temperature.

Custodian:
Air Force - 85

Review activity:
Air Force - 80

Preparing activity:
Air Force - 85

(Project 5910-F272)

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APPENDIX

QUALIFICATION-SUBMITTAL PLANS

10. SCOPE

10.1 This appendix details the qualification-submittal plans for capacitors to be subjected to the qualification inspection specified in this specification. These plans specify the sample size, sampling criteria, and limits of coverage for the type of qualification sought.

20. SUBMITTAL PLANS

20.1 Capacitors.

20.1.1 Single type. The following details apply:

(a) Sample size - As specified in table I.
(b) Sampling criteria - Sample units of the same style, rated voltage, capacitance, capacitance tolerance, and electrolyte.
(c) Limits of coverage - Qualification coverage will be extended to all lower capacitance values and all wider capacitance tolerances than those in the qualified sample, but will not be extended to another style, rated voltage, or electrolyte.

20.1.2 Voltage-group type. The following details apply:

(a) Sample size - As specified in table VII.
(b) Sampling criteria - Sample units of (1) the same style, voltage group, capacitance tolerance, and electrolyte, and (2) the highest capacitance value for the lowest rated voltage and sample units of the highest capacitance value for the highest rated voltage within a single voltage group.
(c) Limits of coverage - Qualification coverage will be extended to all intermediate (or lower) voltages in the voltage group, all lower capacitance values, and all wider capacitance tolerances than those in the qualified sample, but will not be extended to another style, voltage group, or electrolyte.

TABLE VII. Voltage-group submittal data.

Style	Part number	Sample size	Rated voltage	Voltage group
CRL01	M83500/01-0001	18	6	I
	M83500/01-0015	18	30	
	M83500/01-0021	18	50	
	M83500/01-0033	18	125	II
CRL02	M83500/01-1001	18	6	I
	M83500/01-1015	18	30	
	M83500/01-1021	18	50	II
	M83500/01-1033	18	125	

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20.1.3 Combined-style type. The following details apply:

(a) Sample size - As specified in table VIII.
(b) Sampling criteria - Sample units of (1) the same voltage group and electrolyte, and (2) the combination of styles as specified in table VIII; one style of the highest capacitance value for the lowest rated voltage, and the other style of the highest capacitance value for the highest rated voltage within a single voltage group.
(c) Limits of coverage - As specified in 20.1.2(c), except that coverage will be extended to both styles.

TABLE VIII. Combined-style submittal data.

Style	Part number	Sample size	Rated voltage
CRL01 and CRL02	M83500/01-0001-1001	18	6
	M83500/01-0015-1015	18	30
	M83500/01-0021-1021	18	50
	M83500/01-0033-1033	18	125

30. DATA REQUIREMENTS

30.1 Description of items. The supplier shall submit a detailed description of the capacitors being subjected to qualification inspection, including, as applicable, information on the type of electrodes; type of electrolyte; type of seal; the case and lead materials; type of welds; and the case finish.

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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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